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Ohm's Law

Description

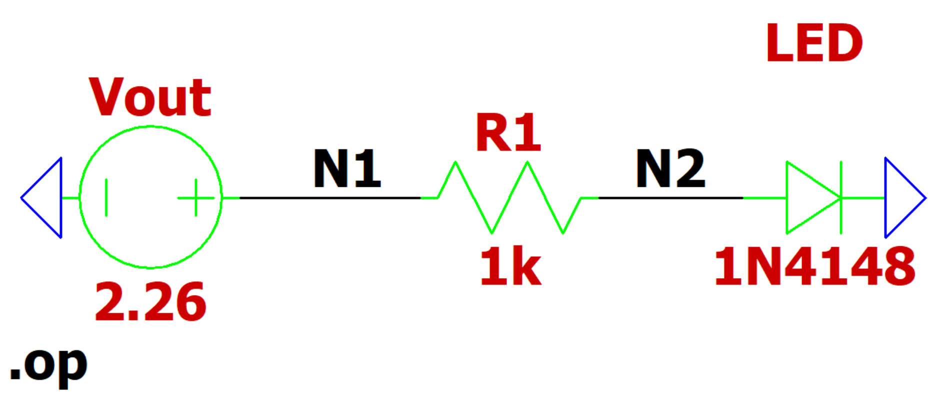
What building block are you applying this concept to?

LED

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

We will use Ohm's Law to derive the resistance we need to obtain ~1 mA (sufficient current to illuminate the LED without overloading the op-amp output) for a 2.26V source (originally 3 V but drops to 2.26 V due to the Op-Amp not being rail-to-rail). We chose ~1 mA because it allows the op-amp to supply the correct voltage to the resistor and LED. If we chose a higher current, the voltage supplied would end up being lower than intended.

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all

variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

I(circuit) = (V(N1) - V(N2)) / R1

R1 = (V(N1) - V(N2)) / I(circuit)

Assuming ideal 3V Vout of op-amp:

R1 = (3V - 2V) / 1mA

R1 = 1k ohm

Assuming 2.26V Vout due to the op-amp's inability to reach the full rail voltage:

I(circuit) = (2.26V - 1.79V) / 1k ohm

I(circuit) = 0.47 mA

Calculate a numerical value for your circuit variable(s).

We calculated that we need roughly a 1k Ohm resistor to obtain ~1 mA. This is due to the 2 V drop from the LED. This would mean that we need roughly a 1 V drop from the resistor, which would be need to be 1k Ohms at ~1 mA current.

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

We will measure the current through the 1k Ohm resistor to confirm that we actually obtain ~1 mA.

Simulation

Make sure that your circuit is simulated in isolation (not connected to any other circuits).

Describe your simulation in 1 to 2 sentences.

Our simulation demonstrates the voltage drop across the resistor that is in series with the LED of our circuit. The schematic only include the necessary components to do so, which is the Vout of the op-amp, the 1k ohm resistor, and the LED.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., '.tran 1m.').

The LED was modeled using a 1N4148 silicon diode in LTSpice as no LED model was available in the library. The 1N4148 has a forward voltage of ~0.6V compared to the measured LED forward voltage of 1.79V, which accounts for the difference between simulated and measured values. The simulated current was 1.65mA compared to the measured 0.47mA.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

```
       --- Operating Point ---

V(n1):          2.26            voltage
V(n2):          0.607862        voltage
I(Led):         0.00165214      device_current
I(R1):          -0.00165214     device_current
I(Vout):        -0.00165214     device_current
```

These values show the nodal voltages and the currents across the circuit simulated in LTspice.

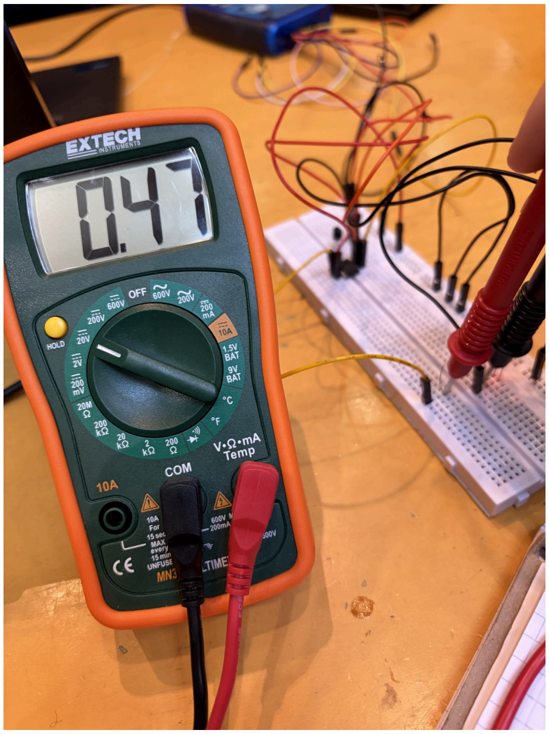
Experimental

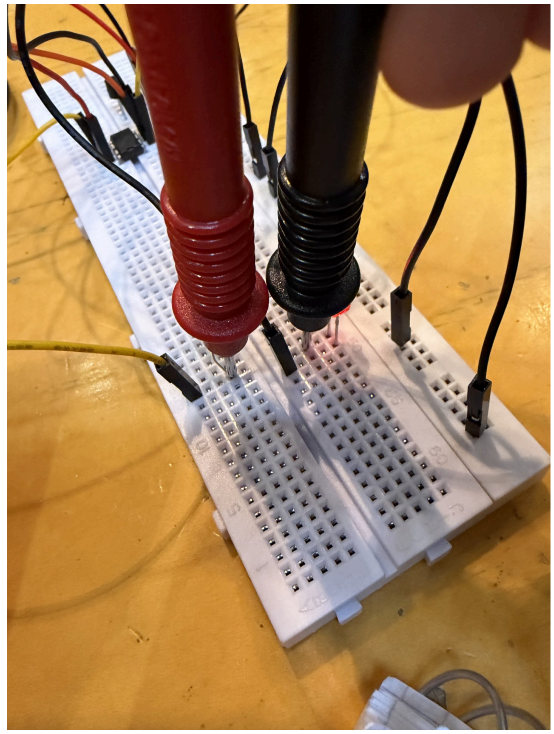
Make sure that your circuit is measured in isolation (not connected to any other circuits).

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

We placed a 1k Ohm resistor in series with a red LED. We measured the current by measuring the voltage across the 1k Ohm resistor and dividing by the resistance. The power was being supplied to the resistor from the Vout terminal of the the OP97.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.





These images show the voltage drop across the resistor. With a voltage drop of 0.47 V, and a resistor of 1k Ohm, the resultant current through the circuit is 0.47 mA, which is just enough to light the LED dimly.

Discussion

Here we interpret out results. If your predicted, simulated, and experimental don't line up, consider re-evaluating your calculations, simulation setup, and experimental setup. Consider possible non-idealities present in components. Make sure you are quantitatively comparing results rather than qualitatively comparing. That is, instead of saying, 'the results look about the same,' calculate a percent error.

Compare your analytical, simulation, and experimental results. Discuss any differences between them. If there is any difference between calc, sim, and measurement provide a reason as to why (DO NOT say it is from 'human error' or 'incorrect wiring').

Between calculations, simulation, and measurement, there were some slight differences. The differences between the simulation and the other two occurred because the correct LED is not apart of the LTspice library, causing the voltage drop across the LED to be different in the simulation. The percent error calculated between the simulation current and the measured current is: (|1.65 - 0.47| / 0.47) * 100 = 251%. The differences between calculation and measurement occur when assuming an ideal op-amp vs. a non-rail-to-rail op-amp. When we assume an ideal op-amp we are assuming that the Vout of the op-amp is 3 V, otherwise we assume that it is 2.26 V like in the measurement. However, if we do assume a Vout of 2.26 V then our measurements and calculations do not differ. The percent error between the calculated and measured current is 0%, confirming that accounting for the OP97's non-rail-to-rail behavior fully explains the deviation from the ideal 1mA.

Discuss how this analysis helped you in your design.

This analysis helped us in our design to find out the correct resistor to use to ensure efficient use of the Vout from the op-amp. It also helped us to figure out the sufficient current to illuminate the LED.

Op Amp as a Comparator

Description

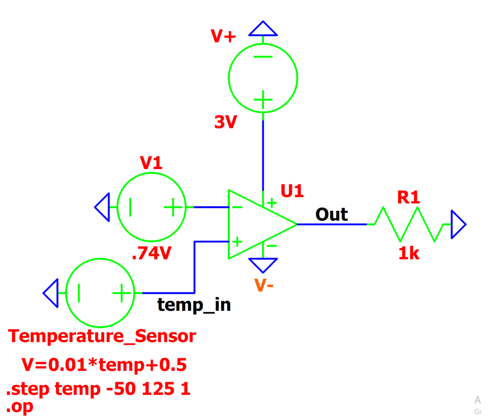
What building block are you applying this concept to?

The temperature threshold trigger to power an LED

What circuit variable(s) will be derived in analysis? Briefly (1 or 2 sentences) describe how you will apply the concept to derive the variable.

We will derive the reference voltagel plugged into the negative terminal of the comparator to trigger the LED at a specific temperature

Paste a well-formatted schematic of the circuit you are analyzing, and include only this circuit and not any others. Label all important nodes; Any nodes or components you discuss in your analysis section should correspond to a labelled node in the schematic.



Analysis

Write the general, symbolic equation(s) you are using to perform your calculation. Please show any lengthy derivations in the appendix to improve readability. Ensure that all variables used in the equation are defined either by labelling them in the schematic or defining them in-text.

V(sensor) = (0.01 Volts)*(1 Degree Celcius) + 0.5 Volts

If V(sensor) > V1, Vout = Vpos

If V(sensor) < V1, Vout = Vneg

Calculate a numerical value for your circuit variable(s).

Given Vpos = 3V and Vneg = 0V(ground)

For a 24 degree Celsius threshold, V1 = 0.01(24)+0.5 = 0.74V

If V(sensor) > 0.74V, Vout = 3V

If V(sensor) < 0.74V, Vout = 0V

State what circuit variable(s) you will measure in your simulation and experimental section to compare with your analysis and confirm intended operation of your circuit.

We will measure the “Out” node relative to the temperature sensor voltage/temperature

Simulation

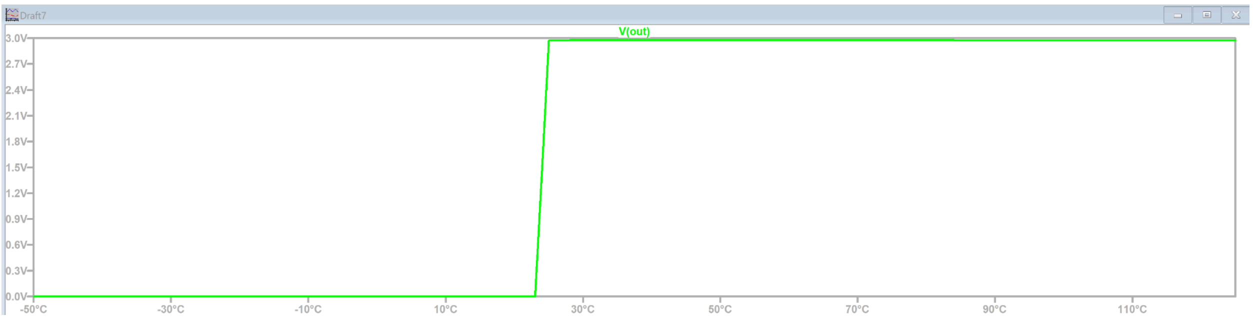
Describe your simulation in 1 to 2 sentences.

Using a .op simulation, we verified that the Vout switches from 0V to 3V exactly when the sensor reached 0.74V(or 24 degrees celsius). The 0 and 3V are given by the V+ and V- terminals on the Op-Amp.

If your simulation schematic differs from your schematic in the description, include it here. Make sure you include all simulation commands (e.g., ‘.tran 1m.’).

The only difference in the simulation is that we had to use a voltage source to simulate the temperature sensor. To do this we set the voltage to be V=.01*temp+0.5 and used the .step command to make it have the temperature range equivalent to the sensor which is -50 to 125 degrees celsius with a step of 1. In the end the function was .step temp -50 125 1.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.

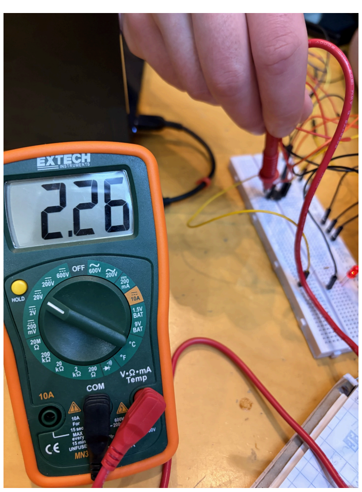


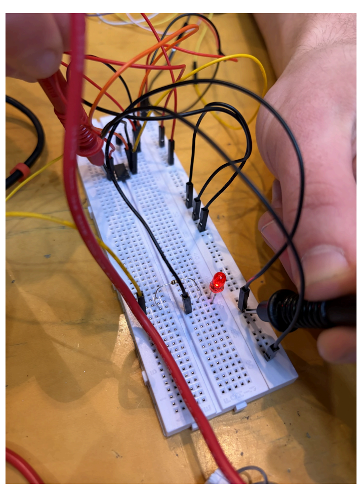
Experimental

Describe your measurement in 1 to 3 sentences. Describe any differences between your analysis/simulation and your experimental circuit. List the part numbers of any components you used outside of passive components (R,L,C), for example which op amp you used.

We used an OP97 Op Amp for our comparator. The comparator takes in the voltage from the TMP36 in IN+, the trigger voltage in IN-, the 3V power in V+, and the 0V ground in V-. We were expecting a Vout of 3V when the comparator was triggered(it was hot enough), but we read a voltage of 2.26 Volts.

Place your simulation results here. This can be a plot, a numerical value, or a table. Make sure your data is formatted well and easy to read. Include all axes on plots and make sure it is clear what value is being measured.





Discussion

Compare your analytical, simulation, and experimental results. Discuss any differences between them. If there is any difference between calc, sim, and measurement provide a reason as to why (DO NOT say it is from ’human error’ or ‘incorrect wiring’).

In the circuit we used an OP97 Op Amp, but in the simulation we used the Universal Op Amp 2. In the simulation we have an ideal Op Amp with no voltage drop, but in reality if the Op Amp is not rail to rail, it cannot produce the exact voltage supplied. The OP97 we used is not rail to rail, so it cannot produce exactly 3V, so the actual voltage was 2.26 V. This means the percent error is -24.7%.

Discuss how this analysis helped you in your design.

This was very crucial information to know going forward. In the future for MS2 and MS3 we may consider getting rail to rail Op Amps, or we can account for this voltage drop in the circuit.